

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Physics

2. **Subject Name:** Electrodynamics **L-T-P:** 3-1-0 **Credits:** 4

3. **Pre-requisites:** None

4. **Syllabus and reference books:**

Syllabus:

Review of Maxwell equations, Electromagnetic waves, Boundary conditions, Fresnel equations, Waveguides, TE and TM modes of rectangular and circular waveguides. (8)
Optical dispersion in materials, resonant absorption, anomalous dispersion, spreading of pulse in dispersive media, Kramers-Kronig relations. (8)

Retarded potentials, Lienard-Wiechert potentials; Radiation from oscillating dipoles and antennas. (6)

Four-vectors; electromagnetic field tensor; transformation of E and B in inertial frames; homogeneous Maxwell equations; Lorentz invariants; Wigner rotation and Thomas precession. Lagrangian formulation of the free Maxwell field; stress-energy-momentum tensor; conserved quantities; inhomogeneous, Maxwell equations; 4-dimensional Green's function of the wave equation. Radiation from accelerated charges; Larmor formula; polar plots and polarization charts; radiation from relativistic motion of charges; linear accelerator and synchrotron radiation; multi-pole expansion. (20)

Reference Books:

1. Classical theory of fields: L. D. Landau and E. M. Lifshitz
2. Introduction to Electrodynamics: D. J. Griffiths
3. Classical Electricity and Magnetism: W.K.H. Panofsky and M. Phillips
4. Electrodynamics: J. D. Jackson

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Review of Maxwell equations, Electromagnetic waves, Boundary conditions, Fresnel equations, Waveguides, TE and TM modes of rectangular and circular waveguides.	8
2.	Optical dispersion in materials, resonant absorption, anomalous dispersion, spreading of pulse in dispersive media, Kramers-Kronig relations.	8
3.	Retarded potentials, Lienard-Wiechert potentials; Radiation from oscillating dipoles and antennas.	6
4.	Four-vectors; electromagnetic field tensor; transformation of E and B in inertial frames; homogeneous Maxwell equations; Lorentz invariants; Wigner rotation and Thomas precession. Lagrangian formulation of the free	20

	Maxwell field; stress-energy- momentum tensor; conserved quantities; inhomogeneous, Maxwell equations; 4-dimensional Green's function of the wave equation. Radiation from accelerated charges; Larmor formula; polar plots and polarization charts; radiation from relativistic motion of charges; linear accelerator and synchrotron radiation; multi-pole expansion.	
Total number of hours		42